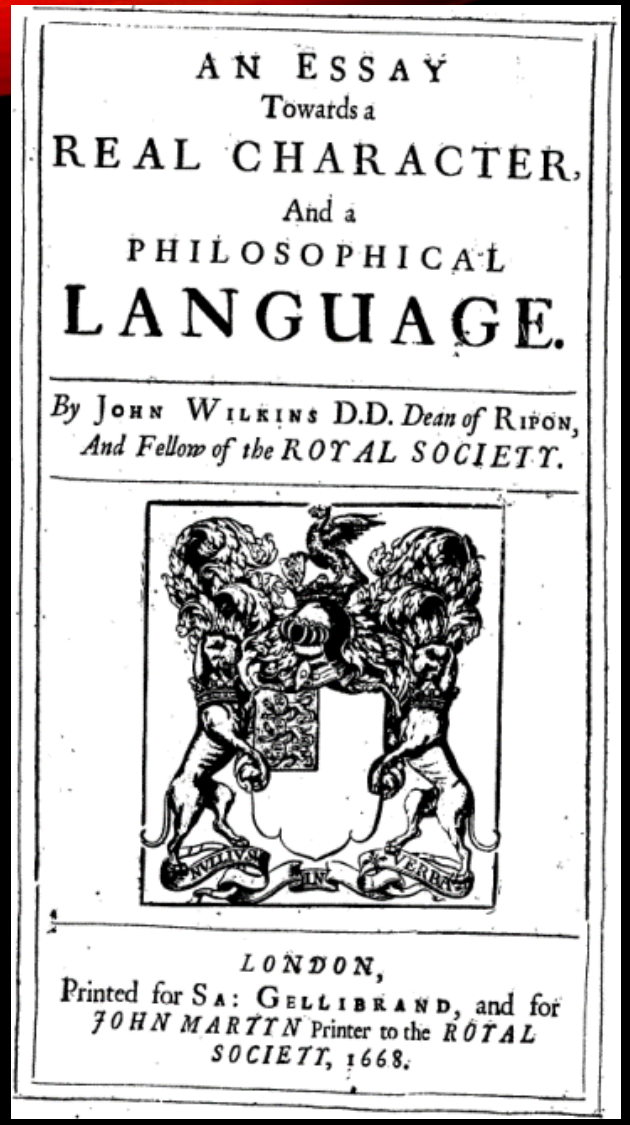
JOHN WILKINS SPEAKS: A DIGITIZATION OF THE *ESSAY TOWARDS A REAL CHARACTER, AND A PHILOSOPHICAL LANGUAGE*

Aidan Wakely-Mulroney

"On Digital Pasts and Digital Futures," California State University Long Beach, April 2023



AN ESSAY
Towards a
REAL CHARACTER,
And a
PHILOSOPHICAL
LANGUAGE.

By JOHN WILKINS D.D. Dean of RIPON,
And Fellow of the ROYAL SOCIETY.

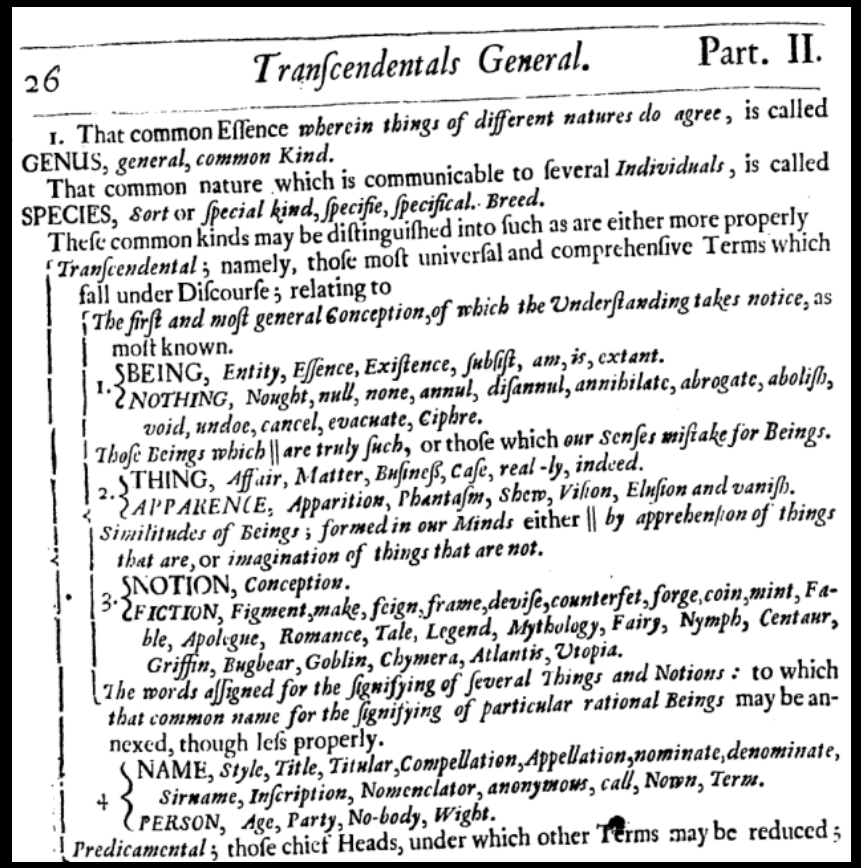
LONDON,
Printed for SA: GELLIBRAND, and for
JOHN MARTIN Printer to the ROYAL
SOCIETY, 1668.

PREPARING THE *ESSAY*

- John Wilkins (1614-72)
  - Dean of Ripon, later Bishop of Chester
  - Fellow of the Royal Society
- Begins his ambitious project ca. 1662
  - Classification of all knowledge
  - Development of a new language
  - Development of new written characters
- The *Essay* finally appears May 4, 1668
  - But his proposals are not adopted

A TEXT VAST IN SCOPE

From the transcendental…



26 *Transcendentals General.* Part. II.

1. That common Essence *wherein things of different natures do agree*, is called GENUS, *general, common Kind.*

That common nature which is communicable to several *Individuals*, is called SPECIES, *Sort* or *special kind, specifie, specifical. Breed.*

These common kinds may be distinguished into such as are either more properly *Transcendental*; namely, those most universal and comprehensive Terms which fall under Discourse; relating to

The first and most general Conception, of which the Understanding takes notice, as most known.

1. BEING, *Entity, Essence, Existence, subsist, am, is, extant.*
   NOTHING, *Nought, null, none, annul, disannul, annihilate, abrogate, abolish, void, undoe, cancel, evacuate, Ciphre.*

Those Beings which || *are truly such*, or those which *our Senses mistake for Beings.*

2. THING, *Affair, Matter, Business, Case, real-ly, indeed.*
   APPARENCE, *Apparition, Phantasm, Shew, Vision, Elusion and vanish.*

Similitudes of Beings; formed in our Minds either || *by apprehension of things that are*, or *imagination of things that are not.*

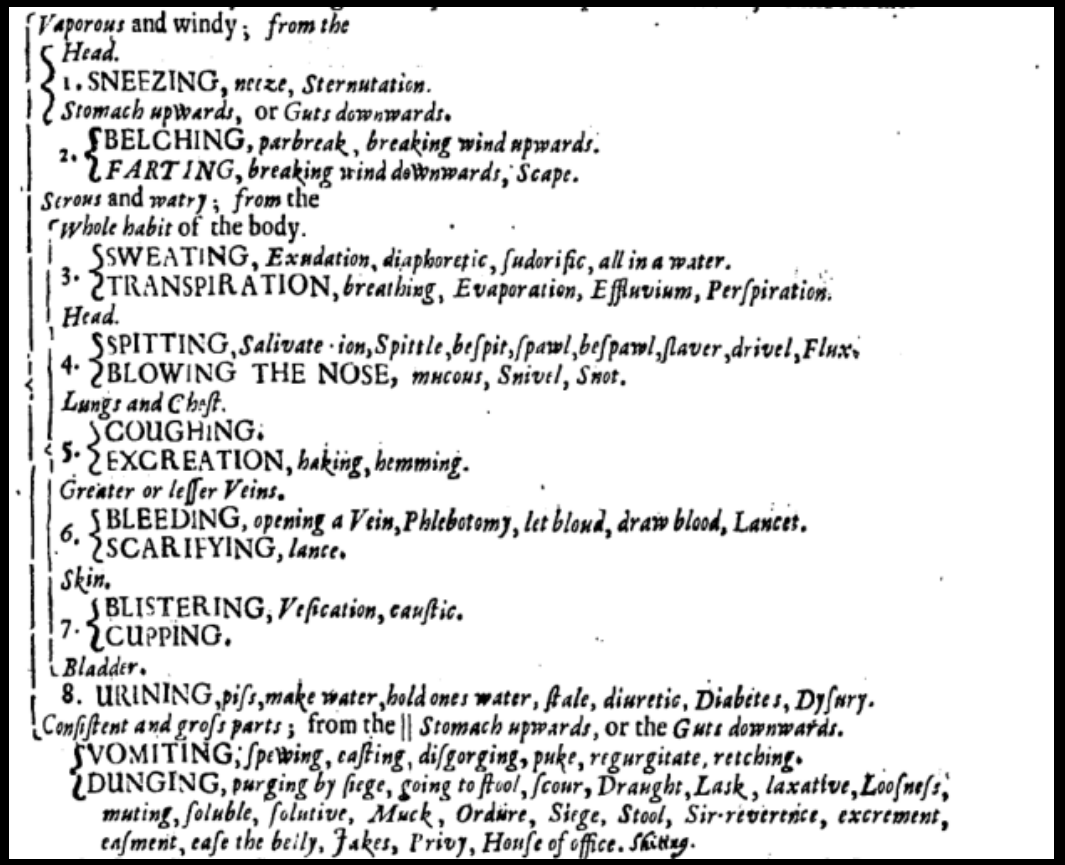
3. NOTION, *Conception.*
   FICTION, *Figment, make, feign, frame, devise, counterfet, forge, coin, mint, Fable, Apologue, Romance, Tale, Legend, Mythology, Fairy, Nymph, Centaur, Griffin, Bugbear, Goblin, Chymera, Atlantis, Utopia.*

The words assigned for the signifying of several Things and Notions: to which *that common name for the signifying of particular rational Beings* may be annexed, though less properly.

4. NAME, *Style, Title, Titular, Compellation, Appellation, nominate, denominate, Sirname, Inscription, Nomenclator, anonymous, call, Nown, Term.*
   PERSON, *Age, Party, No-body, Wight.*

Predicamental; those chief Heads, under which other Terms may be reduced;

…to the profane



Vaporous and windy; *from the*

Head.

1. SNEEZING, *neeze, Sternutation.*

Stomach upwards, or *Guts downwards.*

2. BELCHING, *parbreak, breaking wind upwards.*
   FARTING, *breaking wind downwards, Scape.*

Serous and *watry*; *from* the

Whole habit of the body.

3. SWEATING, *Exudation, diaphoretic, sudorific, all in a water.*
   TRANSPIRATION, *breathing, Evaporation, Effluvium, Perspiration.*

Head.

4. SPITTING, *Salivate-ion, Spittle, bespit, spawl, bespawl, slaver, drivel, Flux.*
   BLOWING THE NOSE, *mucous, Snivel, Snot.*

Lungs and Chest.

5. COUGHING.
   EXCREATION, *haking, hemming.*

Greater or lesser Veins.

6. BLEEDING, *opening a Vein, Phlebotomy, let blood, draw blood, Lancet.*
   SCARIFYING, *lance.*

Skin.

7. BLISTERING, *Vesication, caustic.*
   CUPPING.

Bladder.

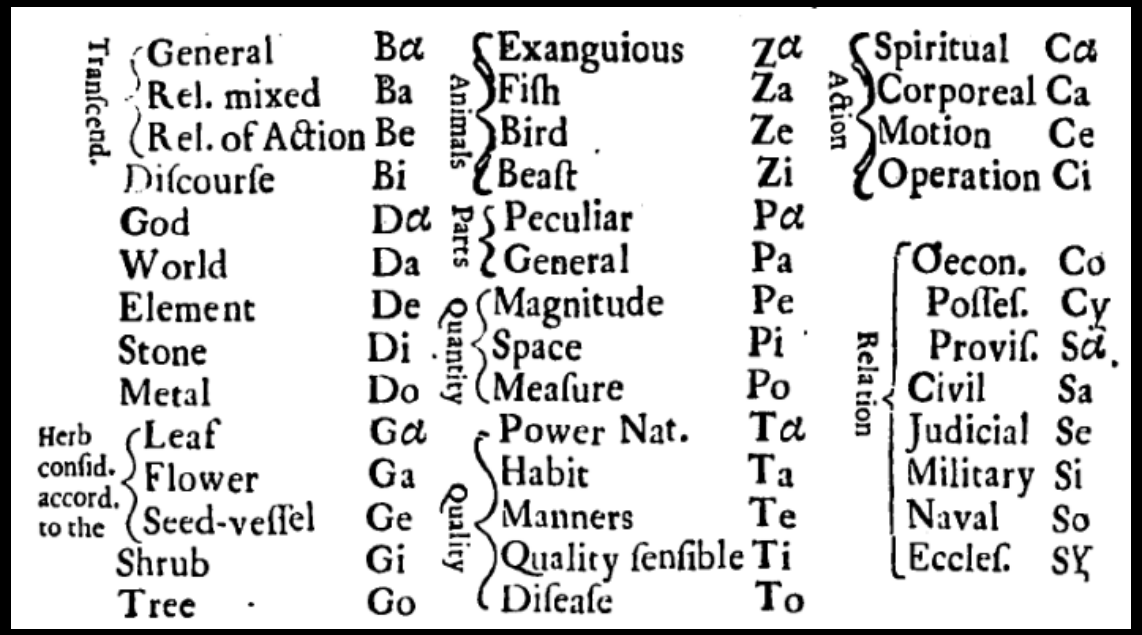
8. URINING, *piss, make water, hold ones water, stale, diuretic, Diabetes, Dysury.*

Consistent and gross parts; from the || *Stomach upwards*, or the *Guts downwards.*

VOMITING, *spewing, casting, disgorging, puke, regurgitate, retching.*
DUNGING, *purging by siege, going to stool, scour, Draught, Lask, laxative, Loosness, muting, soluble, solutive, Muck, Ordure, Siege, Stool, Sir-reverence, excrement, easment, ease the belly, Jakes, Privy, House of office. Skiting.*

WILKINS' CATEGORIZATION SCHEME

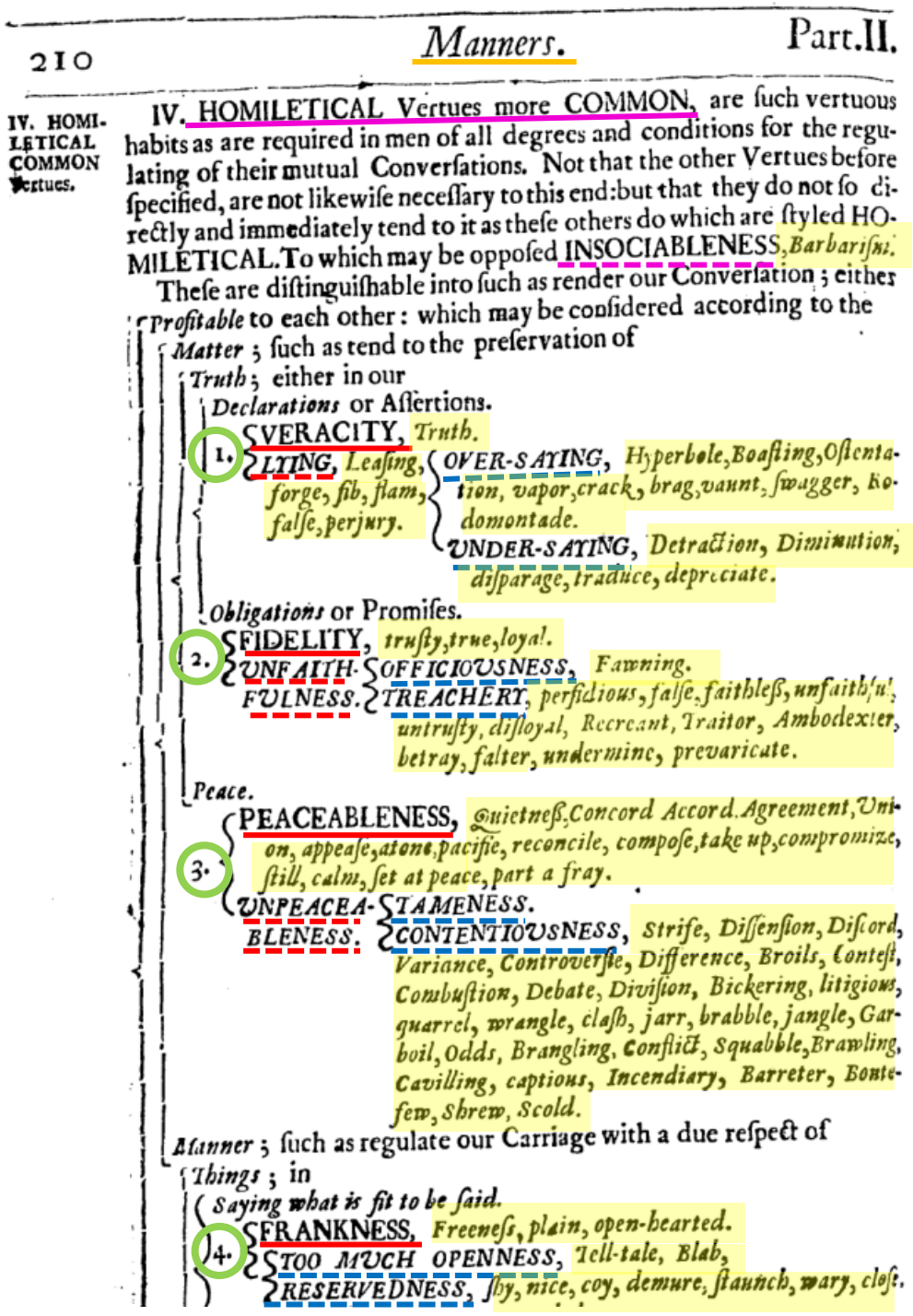
- 40 Genera (shown at right)
  - The top level of organization
- 250 Differences
- 2,036 Species
- 3,823 Radicals
  - The basic unit of communication
  - This includes 1,865 "affinities" and "opposites" to other terms
- +13,000 synonyms
- 266 pages of entries



Transcend.	General	Bα	Animals	Exanguious	Zα	Action	Spiritual	Cα
	Rel. mixed	Ba		Fish	Za		Corporeal	Ca
	Rel. of Action	Be		Bird	Ze		Motion	Ce
	Discourse	Bi		Beast	Zi		Operation	Ci
	God	Dα	Parts	Peculiar	Pα			
	World	Da		General	Pa	Relation	Oecon.	Co
	Element	De	Quantity	Magnitude	Pe		Possef.	Cy
	Stone	Di		Space	Pi		Provif.	Sα
	Metal	Do		Measure	Po		Civil	Sa
Herb confid. accord. to the	Leaf	Gα	Quality	Power Nat.	Tα		Judicial	Se
	Flower	Ga		Habit	Ta		Military	Si
	Seed-vessel	Ge		Manners	Te		Naval	So
	Shrub	Gi		Quality sensible	Ti		Eccles.	SY
	Tree	Go		Disease	To			

EXAMPLE OF THE TIERS

- Genus: "Manners"
- Difference: #4, "Homiletical Vertues more Common"
  - And its opposition, "INSOCIABLENESS"
- Species: 4 are shown here, only given numbers
- Radical: 4 main entries are here, and three paired oppositions (noted in italics); no affinities on this page
  - In #1, "VERACITY" is a main radical and "LYING" is its opposite; in #2
- Double Opposite: 6 here, in brackets beside radicals
  - #4 is a double opposite; #1-#3 are triple opposites
- Synonyms: appear in italics after the main term
  - For the difference, radical, and double opposites



210 *Manners.* Part.II.

IV. HOMILETICAL COMMON Vertues.

IV. HOMILETICAL Vertues more COMMON, are such vertuous habits as are required in men of all degrees and conditions for the regulating of their mutual Conversations. Not that the other Vertues before specified, are not likewise necessary to this end: but that they do not so directly and immediately tend to it as these others do which are styled HOMILETICAL. To which may be opposed INSOCIABLENESS, *Barbarism.*

These are distinguishable into such as render our Conversation; either

Profitable to each other: which may be considered according to the

Matter; such as tend to the preservation of

Truth; either in our

Declarations or Assertions.

1. VERACITY, *Truth.*
   - LYING, *Leasing, forge, fib, flam, false, perjury.*
     - OVER-SAYING, *Hyperbole, Boasting, Ostentation, vapor, crack, brag, vaunt, swagger, Rodomontade.*
     - UNDER-SAYING, *Detraction, Diminution, disparage, traduce, depreciate.*

Obligations or Promises.

2. FIDELITY, *trusty, true, loyal.*
   - UNFAITHFULNESS.
     - OFFICIOUSNESS, *Fawning.*
     - TREACHERY, *perfidious, false, faithless, unfaithful, untrusty, disloyal, Recreant, Traitor, Ambodexter, betray, falter, undermine, prevaricate.*

Peace.

3. PEACEABLENESS, *Quietness, Concord Accord, Agreement, Union, appease, atone, pacifie, reconcile, compose, take up, compromize, still, calm, set at peace, part a fray.*
   - UNPEACEABLENESS.
     - TAMENESS.
     - CONTENTIOUSNESS, *Strife, Dissension, Discord, Variance, Controversie, Difference, Broils, Contest, Combustion, Debate, Division, Bickering, litigious, quarrel, wrangle, clash, jarr, brabble, jangle, Garboil, Odds, Brangling, Conflict, Squabble, Brawling, Cavilling, captious, Incendiary, Barreter, Bontefew, Shrew, Scold.*

Manner; such as regulate our Carriage with a due respect of

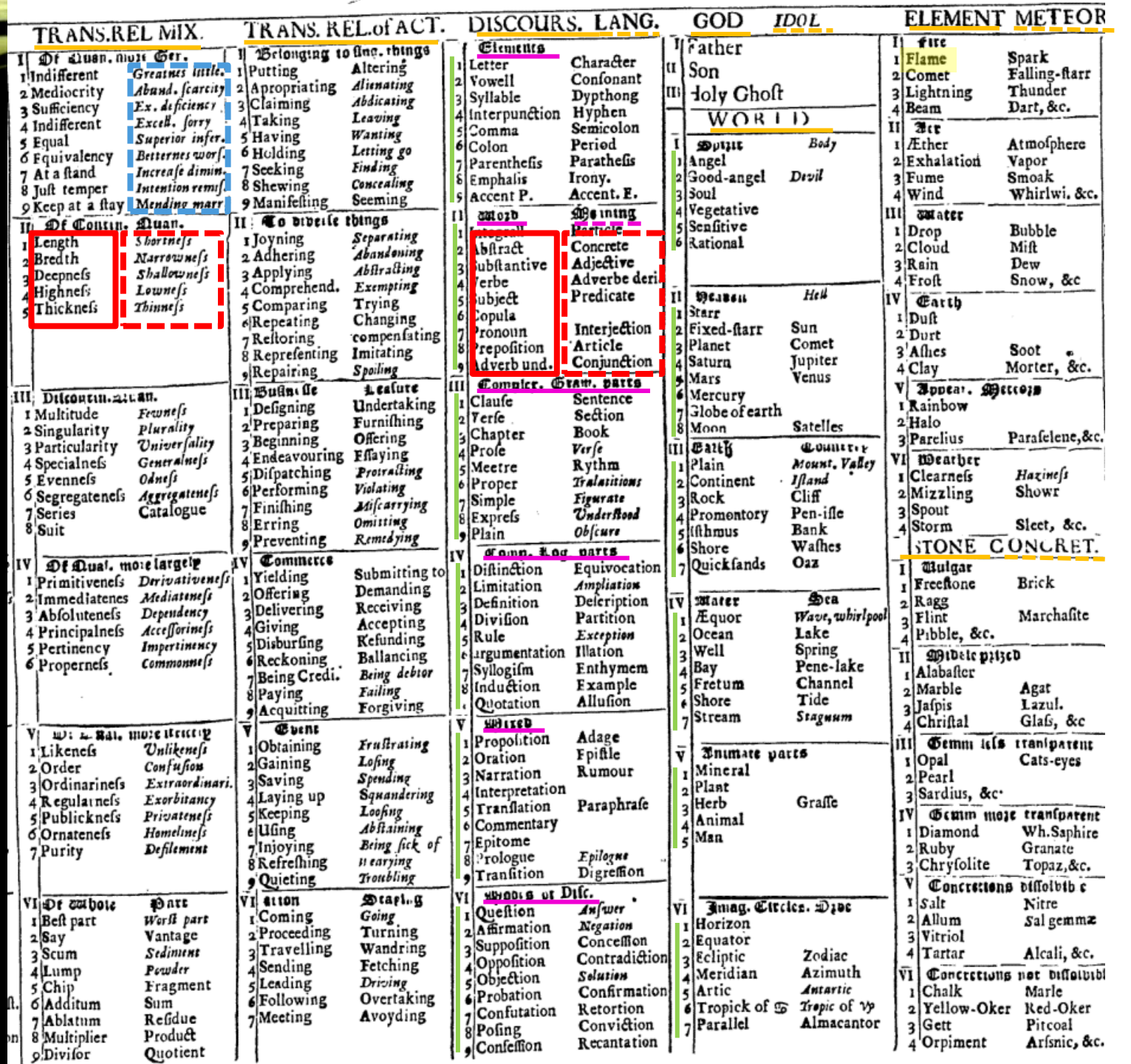
Things; in

Saying what is fit to be said.

4. FRANKNESS, *Freeness, plain, open-hearted.*
   - TOO MUCH OPENNESS, *Tell-tale, Blab,*
   - RESERVEDNESS, *Shy, nice, coy, demure, staunch, wary, close.*

(PART OF) THE SUMMARY TABLE

- Chapter: not indicated
- Genus: column or half column
- Difference: blackletter, with Roman numerals at left
- Species: Arabic numerals
- Radicals: words in regular type
  - Affinities: appear on right side
  - Opposites: appear on right side, in italics; double opps. are separated by commas
- Synonyms: don't appear here



TRANS.REL MIX.

I	Of Quan. more Ger.	
1	Indifferent	*Greatnes little.*
2	Mediocrity	*Abund. scarcity*
3	Sufficiency	*Ex. deficiency*
4	Indifferent	*Excell. sorry*
5	Equal	*Superior infer.*
6	Equivalency	*Betternes worse*
7	At a stand	*Increase dimin.*
8	Just temper	*Intention remiss*
9	Keep at a stay	*Mending marr*
II	Of Contin. Quan.	
1	Length	*Shortnes*
2	Bredth	*Narrownes*
3	Deepnes	*Shallownes*
4	Highnes	*Lownes*
5	Thicknes	*Thinnes*
III	Discontin. Quan.	
1	Multitude	*Fewnes*
2	Singularity	*Plurality*
3	Particularity	*Universality*
4	Specialnes	*Generalnes*
5	Evennes	*Odnes*
6	Segregatenes	*Aggregatenes*
7	Series	Catalogue
8	Suit	
IV	Of Qual. more largely	
1	Primitivenes	*Derivativenes*
2	Immediatenes	*Mediatenes*
3	Absolutenes	*Dependency*
4	Principalnes	*Accessorines*
5	Pertinency	*Impertinency*
6	Propernes	*Commonnes*
V	Of Qual. more Specially	
1	Likenes	*Unlikenes*
2	Order	*Confusion*
3	Ordinarines	*Extraordinari.*
4	Regularnes	*Exorbitancy*
5	Publicknes	*Privatenes*
6	Ornatenes	*Homelines*
7	Purity	*Defilement*
VI	Of Whole	Part
1	Best part	*Worst part*
2	Say	Vantage
3	Scum	*Sediment*
4	Lump	*Powder*
5	Chip	Fragment
6	Additum	Sum
7	Ablatum	Residue
8	Multiplier	Product
9	Divisor	Quotient

TRANS. REL. of ACT.

I	Belonging to sing. things	
1	Putting	Altering
2	Apropriating	*Alienating*
3	Claiming	*Abdicating*
4	Taking	*Leaving*
5	Having	*Wanting*
6	Holding	*Letting go*
7	Seeking	*Finding*
8	Shewing	*Concealing*
9	Manifesting	Seeming
II	To diverse things	
1	Joyning	*Separating*
2	Adhering	*Abandoning*
3	Applying	*Abstracting*
4	Comprehend.	*Exempting*
5	Comparing	Trying
6	Repeating	Changing
7	Restoring	compensating
8	Representing	Imitating
9	Repairing	*Spoiling*
III	Business	Leisure
1	Designing	Undertaking
2	Preparing	Furnishing
3	Beginning	Offering
4	Endeavouring	Essaying
5	Dispatching	*Protracting*
6	Performing	*Violating*
7	Finishing	*Miscarrying*
8	Erring	*Omitting*
9	Preventing	*Remedying*
IV	Commerce	
1	Yielding	Submitting to
2	Offering	Demanding
3	Delivering	Receiving
4	Giving	Accepting
5	Disbursing	Refunding
6	Reckoning	Ballancing
7	Being Credi.	*Being debtor*
8	Paying	*Failing*
9	Acquitting	Forgiving
V	Event	
1	Obtaining	*Frustrating*
2	Gaining	*Losing*
3	Saving	*Spending*
4	Laying up	*Squandering*
5	Keeping	*Loosing*
6	Using	*Abstaining*
7	Injoying	*Being sick of*
8	Refreshing	*Wearying*
9	Quieting	*Troubling*
VI	Action	Stopping
1	Coming	*Going*
2	Proceeding	Turning
3	Travelling	Wandring
4	Sending	Fetching
5	Leading	*Driving*
6	Following	Overtaking
7	Meeting	Avoyding

DISCOURS. LANG.

I	Elements	
1	Letter	Character
2	Vowell	Consonant
3	Syllable	Dypthong
4	Interpunction	Hyphen
5	Comma	Semicolon
6	Colon	Period
7	Parenthesis	Parathesis
8	Emphalis	Irony.
9	Accent P.	Accent. E.
II	Word	Meaning
1	Integral	Partial
2	Abstract	Concrete
3	Substantive	Adjective
4	Verbe	Adverbe deri.
5	Subject	Predicate
6	Copula	
7	Pronoun	Interjection
8	Preposition	Article
9	Adverb und.	Conjunction
III	Complex. Gram. parts	
1	Clause	Sentence
2	Verse	Section
3	Chapter	Book
4	Prose	*Verse*
5	Meetre	Rythm
6	Proper	*Translations*
7	Simple	*Figurate*
8	Expres	*Understood*
9	Plain	*Obscure*
IV	Comp. Log. parts	
1	Distinction	Equivocation
2	Limitation	*Ampliation*
3	Definition	Deseription
4	Division	Partition
5	Rule	*Exception*
6	Argumentation	Illation
7	Syllogism	Enthymem
8	Induction	Example
9	Quotation	Allusion
V	Mixed	
1	Proposition	Adage
2	Oration	Epistle
3	Narration	Rumour
4	Interpretation	
5	Translation	Paraphrase
6	Commentary	
7	Epitome	
8	Prologue	*Epilogue*
9	Transition	Digression
VI	Modes of Disc.	
1	Question	*Answer*
2	Affirmation	*Negation*
3	Supposition	Concession
4	Opposition	Contradiction
5	Objection	*Solution*
6	Probation	Confirmation
7	Confutation	Retortion
8	Posing	Conviction
9	Confession	Recantation

GOD *IDOL*

I	Father	
II	Son	
III	Holy Ghost	

WORLD

I	Spirit	*Body*
1	Angel	
2	Good-angel	*Devil*
3	Soul	
4	Vegetative	
5	Sensitive	
6	Rational	
II	Heaven	*Hell*
1	Starr	
2	Fixed-starr	Sun
3	Planet	Comet
4	Saturn	Jupiter
5	Mars	Venus
6	Mercury	
7	Globe of earth	
8	Moon	Satelles
III	Earth	Country
1	Plain	*Mount. Valley*
2	Continent	*Island*
3	Rock	Cliff
4	Promontory	Pen-isle
5	Isthmus	Bank
6	Shore	Washes
7	Quicksands	Oaz
IV	Water	Sea
1	Æquor	*Wave, whirlpool*
2	Ocean	Lake
3	Well	Spring
4	Bay	Pene-lake
5	Fretum	Channel
6	Shore	Tide
7	Stream	*Stagnum*
V	Animate parts	
1	Mineral	
2	Plant	
3	Herb	Grasse
4	Animal	
5	Man	
VI	Imag. Circles. Spac	
1	Horizon	
2	Equator	
3	Ecliptic	Zodiac
4	Meridian	Azimuth
5	Artic	*Antartic*
6	Tropick of ♋	*Tropic of ♑*
7	Parallel	Almacantor

ELEMENT METEOR

I	Fire	
1	Flame	Spark
2	Comet	Falling-starr
3	Lightning	Thunder
4	Beam	Dart, &c.
II	Air	
1	Æther	Atmosphere
2	Exhalation	Vapor
3	Fume	Smoak
4	Wind	Whirlwi. &c.
III	Water	
1	Drop	Bubble
2	Cloud	Mist
3	Rain	Dew
4	Frost	Snow, &c
IV	Earth	
1	Dust	
2	Durt	
3	Ashes	Soot
4	Clay	Morter, &c.
V	Appear. Meteors	
1	Rainbow	
2	Halo	
3	Parelius	Paraselene,&c.
VI	Weather	
1	Clearnes	*Haziness*
2	Mizzling	Showr
3	Spout	
4	Storm	Sleet, &c.

STONE CONCRET.

I	Vulgar	
1	Freestone	Brick
2	Ragg	
3	Flint	Marchasite
4	Pibble, &c.	
II	Middle prized	
1	Alabaster	
2	Marble	Agat
3	Jaspis	Lazul.
4	Christal	Glass, &c
III	Gemm less transparent	
1	Opal	Cats-eyes
2	Pearl	
3	Sardius, &c.	
IV	Gemm more transparent	
1	Diamond	Wh.Saphire
2	Ruby	Granate
3	Chrysolite	Topaz,&c.
V	Concretions dissolvible	
1	Salt	Nitre
2	Allum	Sal gemmæ
3	Vitriol	
4	Tartar	Alcali, &c.
VI	Concretions not dissolvibl	
1	Chalk	Marle
2	Yellow-Oker	Red-Oker
3	Gett	Pitcoal
4	Orpiment	Arsnic, &c.

THE LANGUAGE: LATINIZED FORM

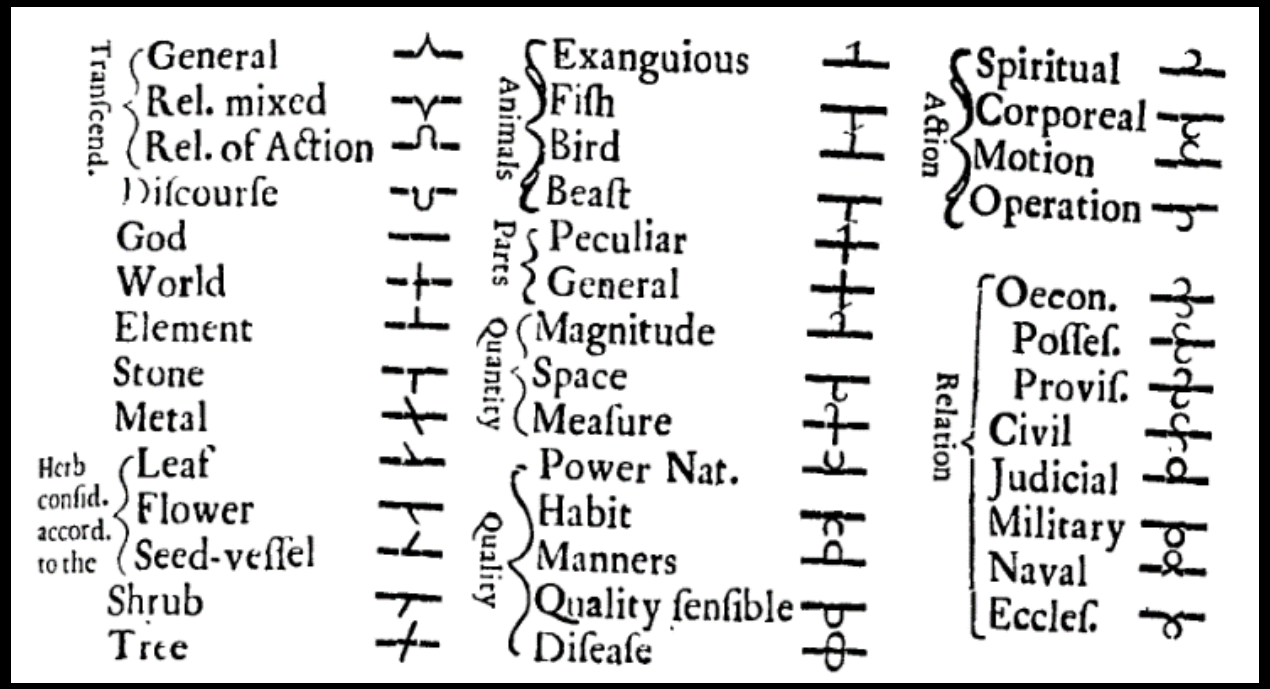
Borges on Wilkins:

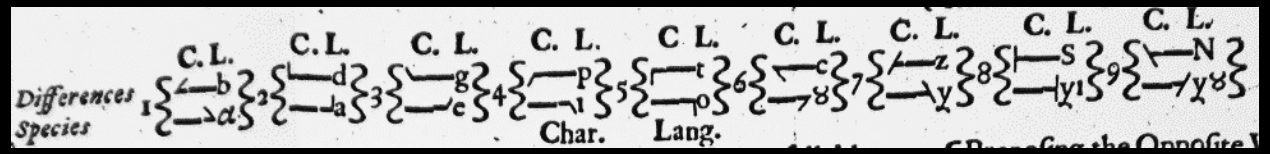
To each class [i.e., genus] he assigned a monosyllable of two letters; to each difference, a consonant; to each species, a vowel.

For example: **de**, *means element;* **deb**, *the first of the elements, fire;* **deba**, *a portion of the element of fire, a flame.*

- Borges' summary is correct, for basic word types
  - And each letter has a specific sound
- Further rules (extra letters and vowels) apply for
  - Opposition/Affinity
  - Substantive/Adjective/Adverb
  - Neutral/Active/Passive
  - Abstract/Concrete
  - Singular/Plural
  - Can differ for genus vs. radical
- Total of 36 different forms
  - The synonyms hint towards these

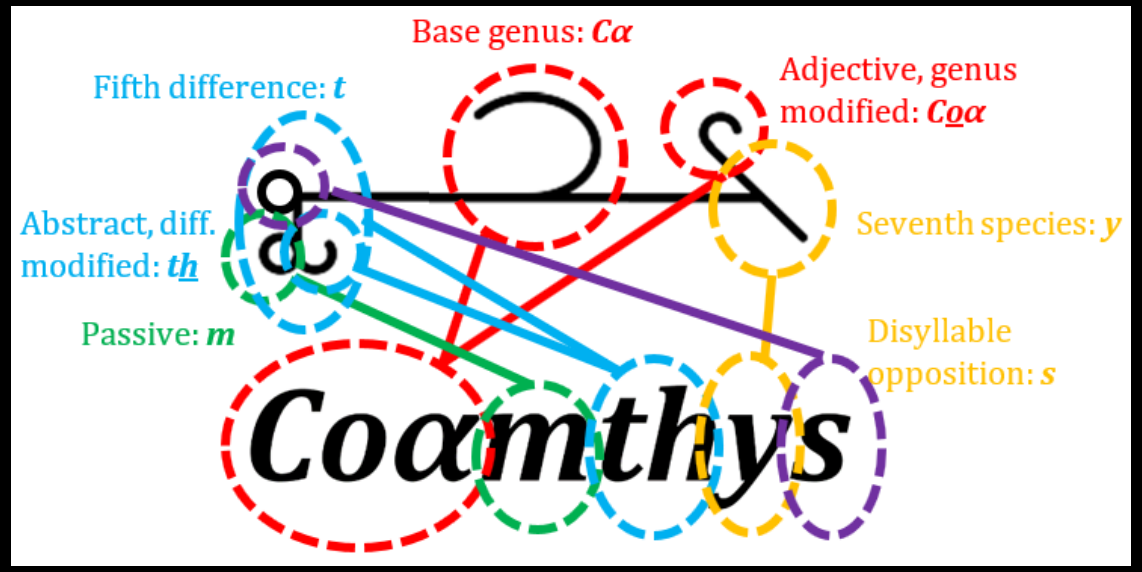
THE CHARACTERS





- Each of the 40 genera has its own unique character (upper box)
  - Differences and radicals are formed by modifying the left and right ends (lower box)
  - Opposition, abstraction, passive, etc. are formed by adding hooks and circles to the ends

LATIN SCRIPT VS. CHARACTERS



- Characters are constructed in a different order than the Latin form
  - The Latinized words are read linearly (segmented script, like English or Arabic)
  - Characters are composed of non-sequential morphemes (logographs, like Chinese characters)
- At left, the abstract, passive, adjective form of "Diffidence"

THE DIGITIZATION

METHODOLOGY

- Goal: a digital version that…
  - Generates Latinized and character forms for every word
  - Provides a complete (and error-free) summary table
  - Is searchable and interactive
- Steps
  - Converting the text into a database (in MS Excel)
  - Adding in modern spellings, to assist modern users in finding words
    - E.g., “Economical” for “Oeconomical”
  - Generating a digital summary table
  - Coding the language rules
  - Drawing the elements for characters and coding rules to combine them

INSIGHTS FROM THE DIGITIZATION

	Chapter	Genus	Difference	Species	Radical	Total
Main Entries/ Pairs	**(11)**	**40**	**250**	**(2,036)**	**2,036**	**2,326**
Unpaired	(11)	22	190	(2,036)	249	461
Affinity	—	9	38	—	1,321	1,368
Opposite	—	9	20	—	312	341
Double Opposite	—	—	2	—	154	156
Total Base Entries	**—**	**58**	**310**	**—**	**3,823**	**4,191**
Synonyms*	—	243	765	—	12,067	13,075
Total Entries (Original Text)	**(11)**	**301**	**1,075**	**(2,036)**	**15,890**	**17,266**

- Numerical breakdown of word forms
- Identification of incomplete grammatical rules
- Discovery of publication errors
  - Errors in species numbering
  - Errors in sample translations and examples
  - Many differences between Wilkins' summary table and the written text

**Division not always clear; others might arrive at slightly different numbers*

COMPARISON: SUMMARY TABLE

The published table

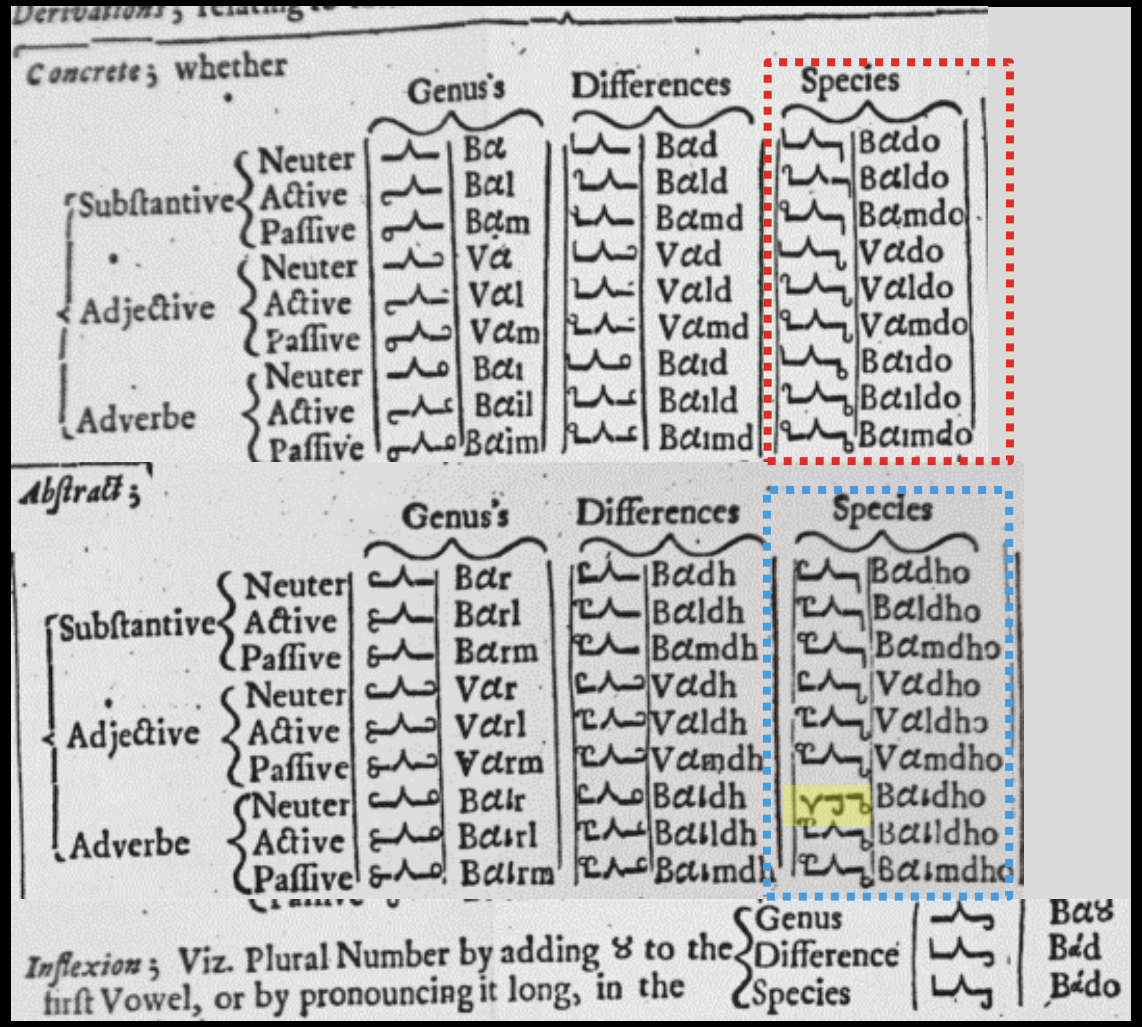
3	Delivering	Receiving	3	Definition	Description	IV	Water	Sea				2	Ragg						
4	Giving	Accepting	4	Division	Partition	1	Æquor	Wave, whirlpool				3	Flint	Marchasite					
5	Disbursing	Refunding	5	Rule	Exception	2	Ocean	Lake				4	Pibble, &c.						
6	Reckoning	Ballancing	6	Argumentation	Illation	3	Well	Spring				II	Middle prized						
7	Being Credi.	Being debtor	7	Syllogism	Enthymem	4	Bay	Pene-lake				1	Alabaster						
8	Paying	Failing	8	Induction	Example	5	Fretum	Channel				2	Marble	Agat					
9	Acquitting	Forgiving		Quotation	Allusion	6	Shore	Tide				3	Jaspis	Lazul.					
V	Event		V	Mixed		7	Stream	Stagnum				4	Christal	Glass, &c					
1	Obtaining	Frustrating	1	Proposition	Adage	V	Animate parts					III	Gemm less transparent						
2	Gaining	Losing	2	Oration	Epistle	1	Mineral					1	Opal	Cats-eyes					
3	Saving	Spending	3	Narration	Rumour	2	Plant					2	Pearl						
4	Laying up	Squandering	4	Interpretation		3	Herb	Grasse				3	Sardius, &c.						
5	Keeping	Loosing	5	Translation	Paraphrase	4	Animal					IV	Gemm more transparent						
6	Using	Abstaining	6	Commentary		5	Man					1	Diamond	Wh.Saphire					
7	Injoying	Being sick of	7	Epitome	Epilogue							2	Ruby	Granate					
8	Refreshing	Wearying	8	Prologue	Digression							3	Chrysolite	Topaz,&c.					
9	Quieting	Troubling	9	Transition		V	Imag. Circles. Zodiac					V	Concretions dissolvible						
VI	Action	Wearing	VI	Words of Disc.		1	Horizon					1	Salt	Nitre					
1	Coming	Going	1	Question	Answer	2	Equator					2	Allum	Sal gemmæ					
2	Proceeding	Turning	2	Affirmation	Negation	3	Ecliptic	Zodiac				3	Vitriol						
3	Travelling	Wandring	3	Supposition	Concession	4	Meridian	Azimuth				4	Tartar	Alcali, &c.					
4	Sending	Fetching	4	Opposition	Contradiction	5	Artic	Antartic				VI	Concretions not dissolvible						
5	Leading	Driving	5	Objection	Solution							1	Chalk	Marle					
	Following	Overtaking	6	Probation	Confirmation														

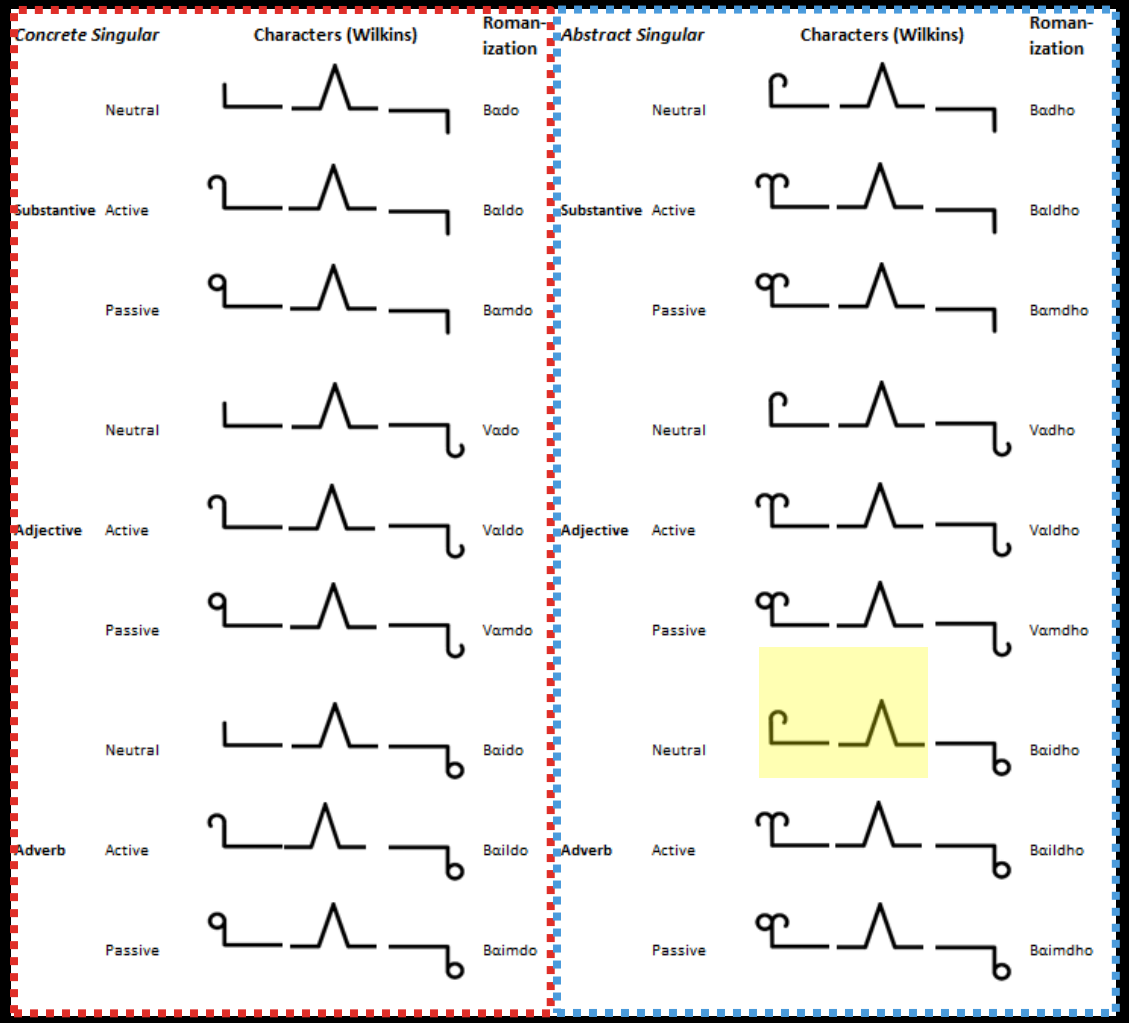
4	Clay	Morter, &c.
V	Appear. Meteors	
1	Rainbow	
2	Halo	
3	Parelius	Paraselene, &c
VI	Weather	

The digital table, noting errors and missing entries

3			4			5			6			7		
V	Event		V	Mixed Notions Of		IV	Water / Sea		V	Appearing Meteors		V	Earthy Concretions	
1	Obteining	*Frustrating*	1	Proposition	Adage	1	Aequor	*Wave Whirl-Pool*	1	Rainbow		1	Salt	Nitre
2	Gaining	*Loosing*	2	Oration	Epistle	2	Ocean	Lake	2	Halo		2	Alume	Sal Gemmae
3	Saving	*Spending*	3	Narration	Rumor	3	Well	Spring	3	Parelius	Paraselene	3	Vitriol	
4	Laying Up	*Squandring*	4	Interpretation		4	Bay	Pene-Lake	4	Virgae		4	Tartar	Alcali
5	Keeping	*Loosing*	5	Translation	Paraphrase	5	Fretum	Channel	5	Chasm		5	Urinous Salt	
6	Using	*Abstaining*	6	Commentary	Epitome	6	Shore	Tide				6	Sal Ammoniac	Chrysocolla
7	Injoying	*Being Sick*	7	Prologue	*Epilogue*	7	Stream	*Stagnum*				7	Sulphur	
8	Refreshing	*Wearying*	8	Transition	Digression							8	Bitumen	Naphtha
9	Quieting	*Troubling*										9	Ambergris	

COMPARISON: WORD FORMS





ASSESSMENT & QUESTIONS

WAS WILKINS A DIGITAL HUMANIST?

- Wilkins used advanced technology
  - The publication of the *Essay* was only possible thanks to advances in typesetting
- Though he lacked a computer, the *Essay* arguably employs computerized techniques
  - Regularized and quantified set of data
  - Elements are manipulated using a specified series of rules and axioms
  - The system is independent of any currently spoken language
- Wilkins is essentially developing a form of analogue code

FURTHER READING

- Lewis, Rhodri. 2012. “The Essay: Wilkins's "Darling".” In *Language, Mind, and Nature: Artificial Languages in England from Bacon to Locke*, by Rhodri Lewis, 146-187. Cambridge: Cambridge University Press.
- Lewis, Rhodri. 2002. “The Publication of John Wilkins's 'Essay' (1668): Some Contextual Considerations.” *Notes and Records of the Royal Society of London* 56 (2): 133-146.
- Markley, Robert. 1993. “Babel Revers'd.” In *Fallen Languages: Crises of Representation in Newtonian England, 1660-1740*, by Robert Markley, 63-94. Ithaca: Cornell University Press.
- Tillery, Denise. 2005. “Engendering the Language of the New Science: The Subject of John Wilkins's Language Project.” *The Eighteenth Century* 46 (1): 59-79.